

TESTING

Testing Guide

Revision: 5 **Last modified:** 2026-08-20T14:17:27Z

This is the **authoritative test-type catalogue** for qBittorrent-Fixed. Every testable module in the repo must have coverage in every applicable row below. The catalogue is derived from Part C of the completion-initiative plan (docs/superpowers/plans/2026-04-19-completion-initiative.md).

Bootstrap (one-time virtualenv setup)

Test dependencies are **not** installed system-wide. Create and populate a local `.venv/` (gitignored — see `.gitignore`) before running anything below:

```
python3 -m venv .venv
.venv/bin/pip install --upgrade pip
.venv/bin/pip install -r tests/requirements.txt
```

Verified working combination (2026-08-08): Python 3.14.6 (`/usr/bin/python3.14`) + `pytest 9.1.1` + `pytest-asyncio 1.4.0` + `fastapi 0.141.1` + `pydantic 2.13.4` + every package listed in `tests/requirements.txt`, all installed fresh into the same virtualenv (no cross-ABI mixing). Re-run the `pip install` line whenever `tests/requirements.txt` changes — it is idempotent.

Once bootstrapped, invoke `pytest` via `.venv/bin/python -m pytest ...` (or activate the venv first with `source .venv/bin/activate` and drop the `.venv/bin/` prefix from every command in this guide).

Test types

#	Type	Framework	Directory	How to run	Coverage report in
1	Unit (Python)	pytest + pytest-asyncio	tests/unit/	pytest tests/unit/ -v --import-mode=importlib	htmlcov/coverage
2	Unit (Frontend)	Vitest + Angular Testing Library	frontend/src/**/*.spec.ts	cd frontend && ng test	frontend/coverage
3	Integration (API)	pytest + httpx.AsyncClient	tests/integration/	pytest tests/integration/ -v --import-mode=importlib	htmlcov/coverage
4	Integration (Plugin ↔ nova3)	pytest + vcrpy (record/replay)	tests/integration/plugins/ (pending Phase 4)	pytest tests/integration/plugins/ -v	htmlcov/
5	End-to-end (browser)	Playwright (Python)	tests/e2e/	pytest tests/e2e/ -v	artifacts/
6	End-to-end (API flow)	pytest + live compose	tests/e2e/ (fixture-gated)	pytest tests/e2e/ -v --import-mode=importlib	htmlcov/
7	Contract (OpenAPI)	schemathesis	tests/contract/	schemathesis run http://localhost:7187/openapi.json	artifacts/contract/
8	Property-based	hypothesis	tests/property/ (pending)	pytest tests/property/ -v	htmlcov/
9	Fuzz	hypothesis + atheris (optional)	tests/fuzz/ (pending)	pytest tests/fuzz/ -v	artifacts/fuzz/
10	Mutation	mutmut	runs against download-proxy/src/ (pending)	mutmut run	html/mutmut/
11	Security	pytest + bandit + semgrep rules	tests/security/	pytest tests/security/ -v	

#	Type	Framework	Directory	How to run	Coverage report in
					htmlcov artifacts/ scans/
12	Performance (latency)	pytest-benchmark	tests/ performance/ tests/ benchmark/	pytest tests/ performance/ tests/benchmark/ -v --benchmark- only	.benchmark
13	Load (throughput)	Locust	tests/load/ (pending Phase 6)	locust -f tests/ load/ locustfile.py	artifacts/ load/
14	Stress (overload)	Locust + chaos probes	tests/stress/	pytest tests/ stress/ -v	artifacts/ stress/
15	Chaos (fault injection)	toxiproxy + pytest	tests/chaos/ (pending)	pytest tests/ chaos/ -v	artifacts/ chaos/
16	Concurrency (race)	pytest-randomly + pytest-repeat + asyncio ops	tests/ concurrency/ (pending)	pytest tests/ concurrency/ -v --count=10	htmlcov
17	Memory leak	tracemalloc + pytest + objgraph	tests/memory/ (pending)	pytest tests/ memory/ -v	artifacts/ memory/
18	Deadlock / timeout	pytest-timeout (hard cap all tests)	global	pytest -v (30 s timeout is the default in pyproject.toml)	test log
19	Smoke	pytest	tests/smoke/ (pending)	pytest tests/ smoke/ -v	htmlcov
20	Monitoring / metrics	pytest + Prometheus scrape asserts	tests/ observability/ (pending)	pytest tests/ observability/ -v	htmlcov
21	Infra / compose	pytest-docker + compose-config validator	tests/infra/ (pending)	pytest tests/ infra/ -v	htmlcov
22	Accessibility	Playwright + axe- core	tests/ally/ (pending)	pytest tests/ ally/ -v	artifacts/ ally/
23	Visual regression	Playwright snapshots	tests/visual/ (pending)	pytest tests/ visual/ --update- snapshots	tests/v __snaps

#	Type	Framework	Directory	How to run	Coverage report in
24	Documentation link check	pytest + linkchecker	tests/docs/ (pending)	pytest tests/docs/ -v	artifacts/docs/
25	Type-check (static)	mypy --strict + tsc --noEmit	CI job	mypy download-proxy/src plugins / cd frontend && tsc --noEmit	CI log
26	Lint (static)	ruff + eslint + shellcheck + hadolint + yamllint	CI job	ruff check . / cd frontend && ng lint / shellcheck *.sh	CI log
27	Dependency audit	pip-audit + npm audit + Snyk + Trivy	CI job	pip-audit -r download-proxy/requirements.txt / snyk test	artifacts/scans/
28	SAST	Semgrep + Bandit + SonarQube	CI job	semgrep scan --config=auto / bandit -r download-proxy/src	artifacts/scans/
29	Secret scan	Gitleaks + TruffleHog	CI job	gitleaks detect	artifacts/scans/
30	License compliance	pip-licenses + license-checker	CI job	pip-licenses / cd frontend && license-checker	artifacts/license/
31	DDoS (abuse resilience)	pytest + Starlette TestClient + httpx ASGI	tests/ddos/	pytest tests/ddos/ -v --import-mode=importlib	htmlcov/ DDoS tests

Rows marked *pending* are scaffolded by the completion-initiative plan phases (see the plan document for the phase that ships each one).

Where tests live

- **Top-level tests/** — all merge-service tests, *not* download-proxy/tests/ (which does not exist).
- **frontend/src/**/*.spec.ts** — co-located Angular unit tests.

- **CI jobs** — `.github/workflows/{syntax,unit,integration,nightly,security}.yaml`.

Service-availability fixtures

Since Phase 0 the following fixtures **start the compose stack** rather than skip when services are down (converted 71 runtime skips to fixture-gated executions, per commit 55d29ce):

Fixture	Starts	Asserts
<code>merge_service_live</code>	<code>qbittorrent-proxy</code>	GET <code>http://localhost:7187/health</code> → 200
<code>qbittorrent_live</code>	<code>qbittorrent</code>	GET <code>http://localhost:7186/</code> → 200
<code>webui_bridge_live</code>	<code>webui-bridge.py</code>	GET <code>http://localhost:7188/health</code> → 200
<code>all_services_live</code>	all three	all of the above

Defined in `tests/fixtures/services.py` and wired into `tests/conftest.py`.

How to add a new test (TDD cadence)

The CLAUDE.md critical constraint is non-negotiable:

1. **RED** — write the failing test first. Name it after the behaviour, not the implementation. Put it in the directory that matches the test type from the catalogue above.
2. **Watch it fail** — run the test in isolation (`pytest tests/unit/test_new_thing.py::test_specific_case -v`) and confirm the failure reason is the one you want (NOT an import error, NOT a fixture error).
3. **GREEN** — write the minimum production code to make the test pass. No refactors yet.
4. **Verify** — run the whole affected test directory to prove nothing else regressed.

5. **Rebuild-reboot** — if the change touched container code, follow the CLAUDE.md REBUILD AND REBOOT constraint:

```
./stop.sh
podman exec qbittorrent-proxy find / -name __pycache__ -type d
      -exec rm -rf {} + || true
./start.sh -p
```

Curl-check the served content matches the committed code.

6. **Commit** — one behaviour per commit. Commit message follows
<type>(phase-N): <subject> when part of a phased plan.

Coverage gate

- pyproject.toml contains [tool.coverage.report] with fail_under set to the current gate (Phase 0 starts low; Phase 5 raises it module-by-module to 100 %).
- The baseline zero point is captured in COVERAGE_BASELINE.md.
- HTML reports land in htmlcov/; XML (for SonarQube) in coverage.xml.

CI entry points

- **Local full pipeline** — ./ci.sh
- **Quick loop** — ./ci.sh --quick
- **Hardcoded-podman runner** — ./run-all-tests.sh (note: hardcodes podman, will fail on docker-only systems — see CLAUDE.md gotcha)
- **Single-file** — ./test.sh --plugin <name> or ./test.sh --full

Per-tracker search stats

Tests that cover TrackerSearchStat and the tracker_started / tracker_completed SSE events live across four suites:

Test file	Type	What it pins
tests/unit/merge_service/ test_tracker_stats.py	Unit	TrackerSearchStat defaults, status transitions (pending → running → success / empty / error / timeout), duration_ms

Test file	Type	What it pins
tests/unit/api_layer/ test_tracker_stats_sse.py	Unit (streaming)	bookkeeping, auth-flag provenance, and sorted to_dict() serialisation. The SSE poll loop emits tracker_started exactly once on pending→running and tracker_completed exactly once per terminal flip; gracefully degrades when metadata predates tracker_stats. sum of results_count == total_results, every completed stat has duration_ms >= 0, set(tracker_stats keys) == set(trackers_searched), failed trackers retain error_type + error.
tests/property/ test_tracker_stats_properties.py	Property (Hypothesis)	POST /api/v1/search, POST /api/v1/search/ sync, and GET /api/v1/ search/{id} all expose the 15-field tracker_stats payload on SearchResponse.
tests/contract/ test_tracker_stats_contract.py	Contract	

The frontend dialog + chip bar are covered by frontend/src/app/components/tracker-stat-dialog/tracker-stat-dialog.component.spec.ts and the tracker stats bar describe block in frontend/src/app/components/dashboard/dashboard.component.spec.ts.

Runtime theme (palette system)

The runtime-switchable palette system (see [docs/DESIGN_SYSTEM.md](#)) is guarded by a four-tier test bundle — every tier asserts *observable* state (DOM, CSS variables, localStorage, Playwright computed style) per the “no false positives” mandate:

Test file	Type	What it pins
tests/unit/ test_palette_catalog.py	Unit (parametric)	Every palette has both variants, every one of the 15 tokens is a valid CSS colour, ids are unique, DEFAULT_PALETTE_ID resolves, Darcula accent is #9d001e. Per-palette parametrisation names the offender.
tests/unit/ test_theme_wiring.py	Unit (integration)	:root fallback block covers every token; dashboard SCSS uses var(--color-*) instead of \$accent/\$bg-* Sass variables; dashboard template includes <app-theme-picker>; TOKEN_CSS_VAR map covers every var.
frontend/src/app/ services/ theme.service.spec.ts	Unit (Vitest)	System-dark detection, stored-state hydration, unknown-id fallback, setPalette writes all 15 tokens with exact values to documentElement.style, toggleMode flips signal + data-mode + color-scheme, prefers-color-scheme change listener respects modeIsUserChosen flag.

Test file	Type	What it pins
frontend/src/app/ components/theme-picker/ theme- picker.component.spec.ts	Unit (Vitest)	One menu item per palette, swatches render with accent/contrast/bg/text colours, clicking an item calls setPalette, clicking outside closes the menu, active palette has .active + aria-checked="true".
tests/e2e/ test_theme_runtime.py	End-to-end (Playwright)	Live dashboard at :7187: picking a new palette changes -- color-accent (+3 others) on <html> to the catalogue value AND writes qbit.theme into localStorage. Skips with a loud message when the served bundle predates the theme-picker.

Whenever the TrackerSearchStat shape changes, run ./scripts/freeze-openapi.sh to refresh docs/api/openapi.json so the contract test test_frozen_and_live_have_same_schemas stays green.

DDoS tests

tests/ddos/ is the §11.4.27(B) / §11.4.169 “**ddos tests**” type: abuse resilience, as distinct from throughput (tests/load/), overload (tests/stress/) and threshold correctness (tests/security/).

What it covers

Dimension	File	The question it answers
Capped burst flood	test_request_flood.py	Are excess requests refused rather than served? Does the refusal path stay clean (no 5xx)?

Dimension	File	The question it answers
Slow client / slow body	test_slow_request.py	Does the service still answer afterwards? Can one flooding client deny service to another client? Does one slow request starve everyone else (head-of-line blocking)? Covers both a slow handler and a dribbled request body.
Payload-size abuse	test_payload_abuse.py	Is an oversized upload refused by an explicit guard (413) rather than buffered and processed? Does an oversized JSON body avoid a 5xx? Can size be used to bypass the per-IP limiter?
Resource-exhaustion admission control	test_resource_exhaustion.py	Does MAX_CONCURRENT_SEARCHES actually engage and shed load with a 429? Does it catch a distributed client whose per-IP budget is untouched?

Every assertion reads a **user-observable outcome** — a status-code population, a response body, a post-attack liveness probe, bystander throughput — never “no exception was raised” (§11.4 / §11.4.1).

Evidence class (§11.4.226): runtime-on-the-real-ASGI-app. The tests drive the real `api.app` through the real middleware stack, real routers and real `slowapi` decorators, **in-process**.

Safety bounds (§12, §12.6, §12.11, CLAUDE.md Mandatory Standard 9)

This host runs live containers and parallel agents, and resource exhaustion here has previously caused forced logouts. The suite is therefore bounded by construction, not by convention. **No bound below may be raised without re-justifying the host budget here.**

Bound	Value	Why
Live stack contact	none	No socket is opened. Nothing ever reaches 127.0.0.1:7187 or any port. A flood against the operator's running stack is forbidden outright.
Largest single burst	MAX_BURST_REQUESTS = 24	~4x the smallest limit under test — enough to prove “refused, not served” with margin. There is no unbounded loop and no “flood until it breaks” anywhere.
Concurrent clients	MAX_CONCURRENT_CLIENTS = 5	Coroutines on ONE event loop. No thread pool, no process pool.
Liveness probes	MAX_LIVENESS_PROBES = 60 @ 20 ms	Sub-millisecond in-process GET /health calls with no I/O.
Largest allocation	OVERSIZE_UPLOAD_BYTES ≈ 10 MiB	Crosses the production 10 MiB guard by 4 KiB — the smallest margin that proves it.
Tracker fan-out	neutralised	SearchOrchestrator._run_search is stubbed, so no test can emit traffic at RuTracker / Kinozal / NNMClub or spawn nova3 subprocesses.
Wall clock	whole suite ≈ 27 s	Well inside the 60 s pytest-timeout cap.

The constants live in tests/ddos/conftest.py and are read by the tests — they are hard caps, not targets.

What this does NOT cover (honest gaps, §11.4.6 / §11.4.3)

Stated so nobody mistakes a green run for more than it is:

1. **TCP-level slowloris is not tested anywhere.** Partial headers, a dribbled TLS handshake, kernel accept-queue exhaustion and uvicorn's own connection / keep-alive timeouts all live *below* ASGI and are unreachable in-process. `test_slow_request.py` covers the ASGI-layer analogue (a dribbled body); the socket-level probe is covered by **neither** this suite nor `challenges/scripts/ddos_resilience_challenge.sh`. Genuinely open.
2. **No multi-host or real-network traffic.** Distributed source addresses are *simulated* via X-Forwarded-For with the production `TRUST_FORWARDED_FOR` knob enabled. Real botnet behaviour, NAT effects and upstream infrastructure are out of scope.
3. **Saturation is simulated, not driven.** `test_resource_exhaustion.py` sets the orchestrator's in-flight counter directly rather than starting real 43-tracker fan-outs — driving true saturation is exactly the off-host damage the host budget forbids. The *guard* under test is real; the *state* is set.
4. **SearchRequest.query has no max_length** (`api/routes.py:187` declares only `min_length=1`). An oversized query is therefore **accepted** and forwarded to the tracker fan-out — a real amplification gap. `test_oversized_query_does_not_5xx_and_service_stays_live` pins the properties that *do* hold (no 5xx, service stays live) rather than asserting a refusal that does not exist. Closing the gap is a production-source change and is **not** done by this suite.
5. **Multi-worker / multi-process limiter coherence is untested.** The default `memory://` slowapi backend is per-process; a deployment scaled past one worker needs `RATE_LIMIT_STORAGE_URI` pointed at Redis for the counters to mean anything. Nothing here exercises that path.

How to run

```
# the suite (in-process, ~27s, safe to run any time)
.venv/bin/python -m pytest tests/ddos/ -v --import-mode=importlib

# the paired §1.1 falsifiability proof – breaks each defence in a
  THROWAWAY
# COPY of download-proxy/src and asserts the suite turns red
bash tests/ddos/mutations/run_mutation_check.sh
```

The mutation runner never touches the real source tree: it copies `download-proxy/src` to a `mktmp -d` directory, mutates the copy, and points the suite at it via the `BOBA_DDOS_SRC_PATH` seam in `tests/ddos/`

conftest.py. This checkout is worked by several agents at once, so an in-place mutation could be swept into someone else's commit before restoration — the §11.4.84 working-tree-quiescence failure mode. It also runs an **unmutated control** first, so a failure caused by the copy mechanism can never be mistaken for a mutation being caught (§11.4.201(1)).

Mutations and the defence each one breaks:

#	Mutation	Must turn red
M1	Rate limits raised to 100000/minute	test_request_flood.py
M2	Limiter key collapsed to a constant (all clients share one bucket)	test_request_flood.py
M3	Upload size guard raised 10 MiB → 10 GiB	test_payload_abuse.py
M4	is_search_queue_full() → always False	test_resource_exhaustion.py
M5	Synchronous time.sleep in the request path (head-of-line blocking)	test_slow_request.py

M5 is why the slow-request oracle is throughput and not latency.

Two earlier versions of that file were bluffs that M5 caught: asserting *ordering* stayed green because a blocking request still finishes last, and asserting per-probe *latency* stayed green because a blocked event loop does not make a bystander probe slow — it stops scheduling it at all (measured: worst probe latency 0.0129 s while only 2 probes ran in a 0.948 s window). Bystander **throughput** is the observable that actually collapses.

Relationship to the neighbouring suites

tests/ddos/ deliberately does not re-test what these already cover:

- tests/unit/test_rate_limit.py — slowapi works, per-IP isolation, 429 body, on a **synthetic** FastAPI app.
- tests/security/test_rate_limit_public_endpoints.py — the **real** app's routes carry the limiter at their **configured thresholds** (§11.4.196(F)).

- `challenges/scripts/ddos_resilience_challenge.sh` — the **live-socket** layer: a bounded localhost concurrency ramp against `:7185 / :7187 / :7189` health surfaces, crash resistance, cross-endpoint isolation, and the BOB-112 `/healthz` TTL-cache regression guard. Requires the stack to be up; this pytest suite does not.

There is one deliberate overlap:

`test_flooding_client_does_not_deny_service_to_another_client` asserts per-client isolation that the unit suite also asserts — but on the **real** app, so it notices a production route or middleware regression that collapses the key function, which a synthetic-app test structurally cannot.

Gotchas

- pytest needs `--import-mode=importlib` for the merge service — the project is not a Python package (no `__init__.py` at the root).
- Integration tests with `requests.get("http://localhost:7187/").ok` guards are **obsolete** since Phase 0 — use the live fixtures instead. `tests/unit/test_no_runtime_service_skips.py` enforces this.
- `pytest-timeout` hard-caps every test at 30 seconds. Stress and load tests that legitimately run longer must override with `@pytest.mark.timeout(...)`.
- The CI workflow is **no longer manual-only** since Phase 0.4 — push and PR triggers the syntax / unit / integration workflows. Only the `nightly.yml` and `security.yml` are scheduled.

Test counts (machine-derived, §11.4.259)

Regenerated by `scripts/compute-badges.sh` — never hand-typed. Corroborates the README badge row + the Contributing-section bullets. Last regenerated: 2026-09-23T16:01:14Z.

Suite	Directory	Real count	Method
Python (whole tests/ tree)	tests/	6195	pytest --collect-only -q
Python unit	tests/unit/	4974	pytest --collect-only -q

Suite	Directory	Real count	Method
Python integration	tests/integration/	308	pytest --collect-only -q
Python e2e	tests/e2e/	35	pytest --collect-only -q
Python contract	tests/contract/	9	pytest --collect-only -q
Frontend (Vitest)	frontend/src/**/*.spec.ts	N/A	vitest list --run produced no parseable test lines
HelixQA Challenges	challenges/scripts/*.sh	38	ls challenges/scripts/*.sh \\ wc -l
Pre-build invariants	scripts/ pre_build_verification.sh	57	max total of every [N/N] progress label

BOB-118 provenance note: the README badge row previously read python tests-585 passing / frontend tests-182 passing with no corroborating source anywhere in this document (a §11.4.6 bluff — the numbers were off by roughly 9x and could not be traced to any real invocation). This table is that corroborating source going forward; wording changed from “passing” to “collected” because collection counts (this table’s method) prove existence, not that every test currently passes — a materially different, honestly-labelled claim.